

: The data requirements is the follows

.An EMPLOYEE is working on a project .1

an employee is identified by an employee ID and has attributes such as -
.employee name, employee position, email, telephone number, and address

.an employee can work on multiple projects -

.a PROJECT is identified by a project ID .2

It has attributes such as project name, project description, project start date, -
.and project end date

The project is worked on by a group of employees. -in project there is group -
.of employee works on it

.The DEPARTMENT consists of work individuals .3

The department is defined by a D_number and has a Dname, contact -
.information

.The department has multiple tasks associated with it -

department have any number of employee , but the one employee works in -
.one department

department control multiple number of project but the project related to one -
.department

.The TASKS are performed by the department members .4

The tasks are defined by a Task-ID and have a task description, task start date, -
.task end date, and task status

.Each member (employee) in the department can handle multiple tasks -

.The RESOURCES are manufactured by the factory .5

The resource is identified by the resource ID and has the resource name, type, -
and information related to it

.Resources are tied to only one factory-

.each project need many resources to complete the work-

The FACTORY makes the resources that workers need to work in .6
construction. - The factory is identified by the factory number in addition to the
.factory name, factory location, and factory information

.Each factory supplies a set of building materials (resources)-

TIME LINE is where the time spent by each team member in performing their . 7
.tasks is recorded

a timeline is identified by a Timeline ID and includes a timeline description, -
.timeline start date, and timeline end date

.Each timeline is associated with a set of reports-

.The timeline is related to one project and the project has one timeline-

.REPORTS are used to document and evaluate the project progress .8

a report is identified by a Report ID and includes additional attributes such as -
.report description and report creation date

.Each report includes a single timeline-

.each report related to one project and the project has many report-

.The CLIENT has contracts with the company .9

A client is identified by client ID and has these attributes: Client telephone-no -
.and customer name

.The client can be linked to more than one contract-

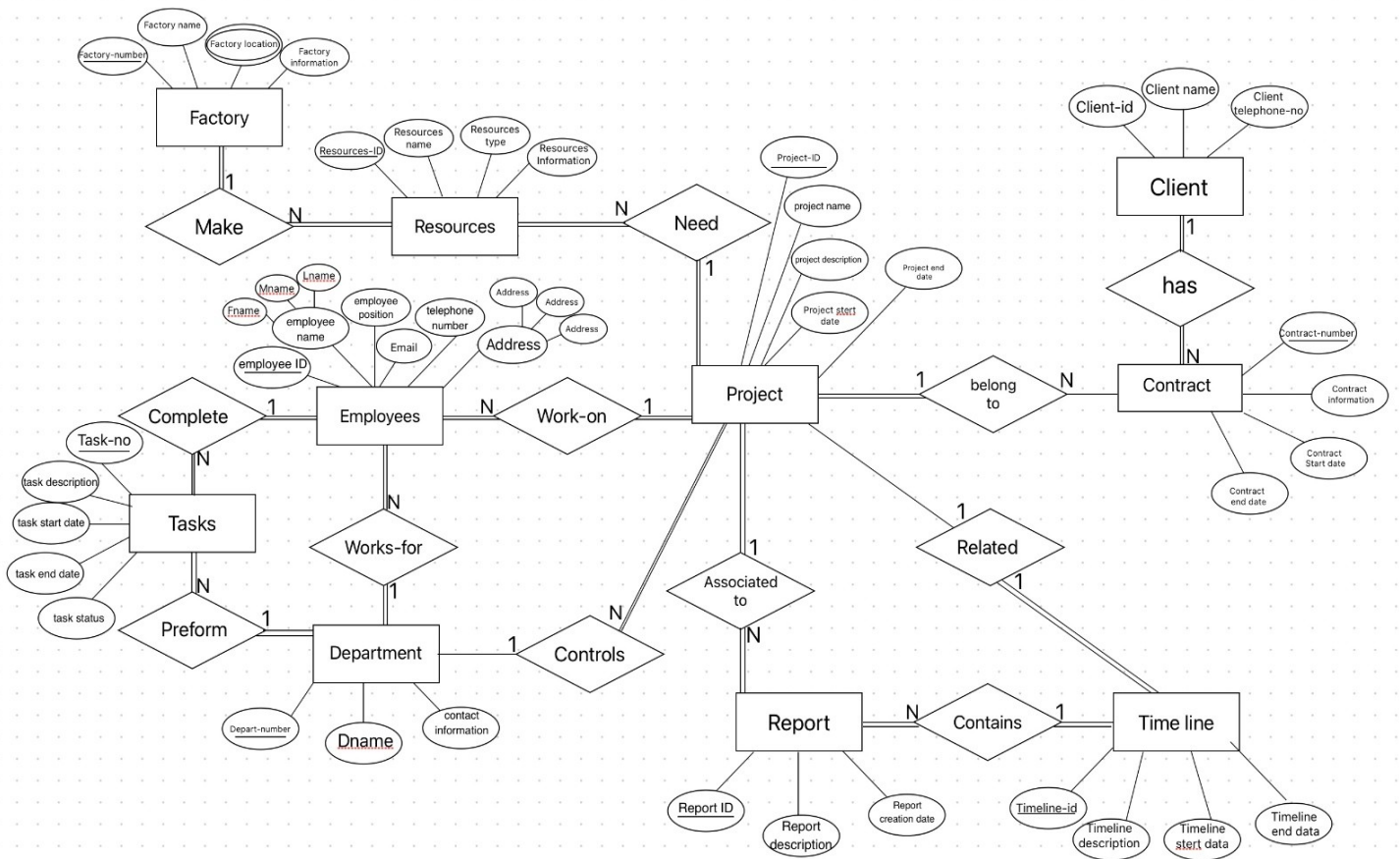
.CONTRACTS are an agreement between the client and the company .10

Contracts identify by contract number, and has contract information, contract -
.start date, contract end date

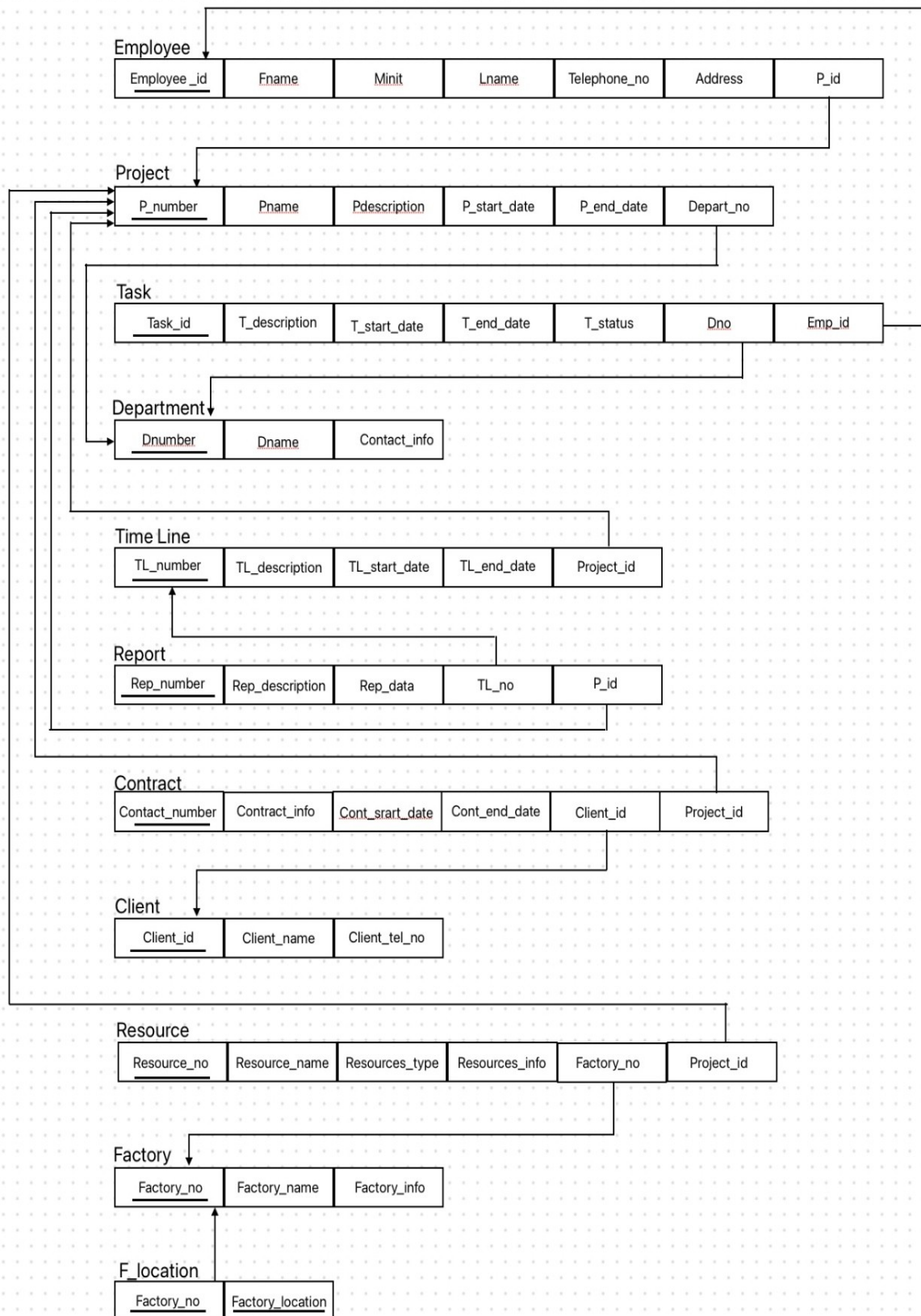
.Contracts relate to one client but The client can has many contracts -

.contract is belong to one project and the project belong to many contract-

:ER Diagram (conceptual data model)



: Relational Data Model/ Relational SCHEMA



:RELATIONAL DATABASE

EMPLOYEE

Emp_ID	fname	lname	position	contacts	project_id
123456789	Gorge	Molten	Anstasia	gorge@gmail.com	104
276543219	Jane	Smith	Developer	Jane.smith@gmail.com	102
345678912	Sami	Ahmad	Ibrahim	Sam3@gamil.com	105
987654312	Davi	Johnso n	Analyst	David.johnson@gamil.com	103
987654321	John	Doe	Manager	John.doe@gmail.com	101

Dnumber	Dname	Contact_info
10	HR	hr@gmail.com
20	Finance	finance@gmail.com
30	Marketing	marketing@gmail.com
40	IT	it@gmail.com
50	Operations	operations@gmail.com

F_LOCATION

Client_ID	Client_name	Client_tel_no
111111111	Sarah Johnson	0198567382
222222222	Daniel Anderson	0183471384
333333333	Olivia Brown	0148254345
444444444	Willim Taylor	0144772298
555555555	Sophia Martinez	0145322768

Factory_on	Flocation
1005	Jazan
1010	Makkah
1015	Jeddah
1020	Abha
1025	Riyadh

Factory_on	Factory_name	Factory_info
1005	Cement factory	This factory produces high- ...quality cement using ad
1010	Steel factory	This factory supplies reliable ...steel products
1015	Processed glass factory	This factory provides insulated ...glass for building
1020	Tile and ceramic factory	This factory offers high-quality ...tiles and ceramic
1025	Gin wood factory	This factory provides environmentally protected wo

Contracts_NUM	Contracts_info	Cont_start_date	Cont_end_date	client_ID	P_id
135792489	Contract for interior design	2022-04-10	2022-11-10	4444444444	104
246813792	Contract for electrical works	2022-03-30	2022-09-30	3333333333	103
789457897	Contract for construction materials	2022-01-01	2022-12-31	1111111111	101
864297532	Contract for landscaping	2022-05-20	2022-10-20	5555555555	105
987657216	Contract for plumbing services	2022-02-15	2022-12-15	2222222222	102

P_number	Pname	Pdescription	Start_date	End_date	Depart_no
101	ProjectM	Cooperative take management system	2023-01-01	2023-04-10	50
102	ProjectQ	Online foreign Language Learning platform	2023-02-15	2023-07-20	40
103	ProjectZ	Digital library management system	2023-03-05	2023-06-15	30
104	ProjectX	Corporate social networking Application	2023-07-20	2023-09-19	20
105	ProjectK	Commercial and entertainment complex	2023-08-12	2023-11-23	20

RESOURCES

R_no	R_name	R_type	resource_information	factory_no	Project_id
5557	Wood metal	Metal	a high-quality	1005	101
8766	iron metal	Metal	Used for specific tasks	1010	102
3255	Oil	Energy resource	It is used for electricity	1015	103
4499	Gas	Natural resource	The item is natural	1020	104
8822	Coal	Natural resource	The item is natural	1025	105

TIMELINE

TL_number	timeline_date	timeline_description	timeline_end_date	Project_id
20	2023-04-10	Installation of plumbing and electrical systems	2023-07-20	104
21	2023-05-20	Final inspection and handover	2023-09-20	105
23	2023-03-30	Completion of foundation work	2023-08-15	103
24	2023-02-15	Commencement of construction	2023-04-20	102
25	2023-01-01	Preparation of project proposal	2023-02-20	101

REPORT

Rep_number	Rep_description	report_creation_date	TL_no	P_id
3232	Construction progress update	2022-02-20	21	102
3344	Initial project assessment	2022-01-10	20	101
5432	Quality inspection report	2022-03-15	23	103
6789	Post-construction evaluation	2022-05-25	25	105
7766	Project completion report	2022-04-30	24	104

CONTRACTS

Contracts_NUM	Contracts_info	Cont_start_date	Cont_end_date	client_ID	P_id
135792489	Contract for interior design	2022-04-10	2022-11-10	4444444444	104
246813792	Contract for electrical works	2022-03-30	2022-09-30	3333333333	103
789457897	Contract for construction materials	2022-01-01	2022-12-31	1111111111	101
864297532	Contract for landscaping	2022-05-20	2022-10-20	5555555555	105
987657216	Contract for plumbing services	2022-02-15	2022-12-15	2222222222	102

Schema DDL (SQL):

codes for COMS

```
CREATE TABLE employee (  
    employee_id    INT                NOT NULL,  
    fname          VARCHAR(255),  
    lname          VARCHAR(255),  
    position       VARCHAR(255),  
    contacts       VARCHAR(255),  
    p_id           INT                NOT NULL,  
    PRIMARY KEY( employee_id),  
    FOREIGN KEY (p_id) REFERENCES projects(p_number)  
);
```

```
CREATE TABLE DEPARTMENT (  
    Dnumber        Int                NOT NULL,  
    Dname          Varchar(15)       NOT NULL,  
    Contract_info  Varchar(30),  
    PRIMARY KEY (Dnumber) );
```

```
CREATE TABLE projects (  
    project_id     INT                NOT NULL,
```

```

P_number      INT      NOT NULL ,
Pname          varchar(15),
Pdescription   varchar(50),
start_date    DATE,
end_date       DATE,
Depart_no     INT      NOT NULL ,
PRIMARY KEY (P_number),
FOREIGN KEY (Depart_no)REFERENCES
DEPARTMENT(Dnumber)
);

```

```

CREATE TABLE TASK (
Task_id        Int          NOT NULL,
T_description   Varchar(60),
T_start_date    date        NOT NULL,
T_end_date      date        NOT NULL,
T_status        Varchar(20) NOT NULL,
Dno             Int          NOT NULL,
Emp_id         Int          NOT NULL,
PRIMARY KEY(Task_id),

```

FOREIGN KEY(Dno) REFERENCES DEPARTMENT
(Dnumber),

FOREIGN KEY (Emp_id) REFERENCES employees
(employee_id)

);

CREATE TABLE Timeline (

TL_number INT NOT NULL,

timeline_date DATE,

timeline_description VARCHAR(255),

timeline_end_date DATE,

project_id INT NOT NULL,

PRIMARY KEY(TL_number),

FOREIGN KEY (project_id)REFERENCES

PROJECT(P_numper)

);

CREATE TABLE Reports (

```

Rep_number          INT          NOT NULL,
Rep_description      VARCHAR(255),
Rep_creation_date    DATE,
TL_NO               INT          NOT NULL,
p_id                INT          NOT NULL,
PRIMARY KEY(Rep_number),
FOREIGN KEY (TL_NO) REFERENCES
timeline(TL_number),
FOREIGN KEY (p_id) REFERENCES projects(P_number)
);

```

```

CREATE TABLE client (
client_ID           INT          NOT NULL,
Client_name         VARCHAR(15),
Client_tel_no       VARCHAR(255),
PRIMARY KEY (client_ID)
);

```

```

CREATE TABLE contracts (

```

```

contracts_NUM          INT          NOT NULL,
contracts_info         VARCHAR(255),
Cont_start_date        DATE,
Cont_end_date          DATE,
client_ID              INT          NOT NULL,
project_id             INT          NOT NULL,
PRIMARY KEY (contracts_NUM),
FOREIGN KEY (client_ID) REFERENCES client(client_ID),
FOREIGN KEY (project_id) REFERENCES
projects(P_number)
);

```

```

CREATE TABLE factory (
factory_no             INT ,
factory_name           VARCHAR(255),
factory_info           VARCHAR(255),
PRIMARY KEY(factory_no)
);

```

```

CREATE TABLE F_location(

```

```
Factory_no      INT          NOT NULL,  
Flocation      VARCHAR(100),  
PRIMARY KEY (Factory_no,Flocation)  
);
```

```
CREATE TABLE resources (  
    Resource_no      INT          NOT NULL,  
    Resource_name    VARCHAR(20),  
    Resource_type    VARCHAR(20),  
    Resource_info    VARCHAR(150),  
    factory_number    INT          NOT NULL,  
    project_ID       INT          NOT NULL,  
    PRIMARY KEY (Resource_no), FOREIGN KEY  
(factory_number) REFERENCES factory(factory_no),  
    FOREIGN KEY (project_ID) REFERENCES  
projects(P_number)  
);
```

THE INSERTS:

INSERT INTO DEPARTMENT

VALUES

(10, 'HR', 'hr@gmail.com'),

(20, 'Finance', 'finance@gmail.com'),

(30, 'Marketing', 'marketing@gmail.com'),

(40, 'IT', 'it@gmail.com'),

(50, 'Operations', 'operations@gmail.com');

INSERT INTO TASK

VALUES

(324, 'Complete report', '2022-01-10', '2022-01-15', 'In progress', 10, 987654321),

(334, 'Prepare presentation', '2022-02-05', '2022-02-10', 'Completed', 20, 2765432198),

(232, 'Research market trends', '2022-03-01', '2022-03-10', 'In progress', 30, 987654312),

(213, 'Develop new feature', '2022-04-15', '2022-04-30', 'Pending', 40, 123456789),

(234, 'Review code quality', '2022-05-20', '2022-05-25', 'Completed', 50, 345678912);

INSERT INTO employee

VALUES

(987654321, 'John', 'Doe', 'Manager',
'john.doe@gmail.com', 5),

(276543219, 'Jane', 'Smith', 'Developer',
'jane.smith@gmail.com', 6),

(987654312, 'Davi', 'Johnson', 'Analyst',
'david.johnson@gmail.com', 7),

(123456789, 'Gorge', 'Molten',
'Anstasia', 'gorge@gmail.com', 8),

(345678912, 'Sami', 'ahmed', 'Ibrahim',
'sam3@gmail.com', 9);

INSERT INTO projects

VALUES

(101, 'projectM', 'Cooperative Task Management
System', '2023-01-01', '2023-04-10', 50),

(102,'projectQ','Online Foreign Language Learning Platform','2023-02-15','2023-07-20',40),

(103,'projectZ','Digital Library Management System','2023-03-05','2023-06-15',30),

(104,'projectX','Corporate Social Networking Application','2023-07-20','2023-09-19',20),

(105,'projectK','Commercial and Entertainment Complex','2023-08-12','2023-11-23',20);

INSERT INTO Timeline

VALUES

(025, '2023-01-01', 'Preparation of project proposal', '2023-02-20',101),

(024, '2023-02-15', 'Commencement of construction work', '2023-04-20',102),

(023, '2023-03-30', 'Completion of foundation work', '2023-08-15',103),

(020, '2023-04-10', 'Installation of plumbing and electrical systems', '2023-07-20',104),

(021, '2023-05-20', 'Final inspection and handover', '2023-09-20',105);

INSERT INTO Reports

VALUES

(3344, 'Initial project assessment', '2022-01-10', 20, 101),
(3232, 'Construction progress update', '2022-02-20', 21, 102),
(5432, 'Quality inspection report', '2022-03-15', 23, 103),
(7766, 'Project completion report', '2022-04-30', 24, 104),
(6789, 'Post-construction evaluation', '2022-05-25', 25, 105);

INSERT INTO client

VALUES

(111111111, 'Sarah Johnson', '0198567382'),
(222222222, 'Daniel Anderson', '0183471384'),
(333333333, 'Olivia Brown', '0148254345'),
(444444444, 'William Taylor', '0144772298'),
(555555555, 'Sophia Martinez', '0145322768');

INSERT INTO contracts

VALUES

(789457897, 'Contract for construction materials', '2022-01-01', '2022-12-31', 111111111, 101),

(987657216, 'Contract for plumbing services', '2022-02-15', '2022-12-15', 222222222, 102),

(246813792, 'Contract for electrical works', '2022-03-30', '2022-09-30', 333333333, 103),

(135792489, 'Contract for interior design', '2022-04-10', '2022-11-10', 444444444, 104),

(864297532, 'Contract for landscaping', '2022-05-20', '2022-10-20', 555555555, 105);

INSERT INTO factory

VALUES

(1005, 'Cement factory', 'This factory produces high-quality cement using advanced techniques.'),

(1010, 'steel factory', 'This factory supplies reliable steel products and components for construction projects.'),

(1015, 'Processed glass factory', 'This factory provides insulated glass for building facades and windows.'),
(1020, 'Tile and ceramic factory', 'This factory offers high-quality tiles and ceramics for building finishes.'),
(1025, 'Gin wood factory', 'This factory provides environmentally protected wood for building structures.'
');

INSERT INTO f_location

VALUES

(1005,'JAZAN'),
(1010,'MAKKAH'),
(1015,'JEDDAH'),
(1020,'ABHA'),
(1025,'RIYADH');

INSERT INTO resources

VALUES

(5557,'Wood metal','Metal','a high quality',1005,101),
(8766,'iron metal','Metal','Used for specific
tasks',1010,102),
(3255,'Oil','Energy resources','it is used for
electricity',1015,103),
(4499,'Gas','Natural resource','the item is natural',
1020,104),
(8822,'Coal','Natural resource','the item is
natural',1025,105);

